SECTION 1 (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated **according to the following marking scheme**:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases.

Q.1 The heating of NH_4NO_2 at 60–70 °C and NH_4NO_3 at 200–250 °C is associated with the formation of nitrogen containing compounds **X** and **Y**, respectively. **X** and **Y**, respectively, are

(A)	N_2 and N_2O
(B)	NH_3 and NO_2
(C)	NO and N_2O
(D)	N_2 and NH_3

Answer: A

Q.2 The correct order of the wavelength maxima of the absorption band in the ultraviolet-visible region for the given complexes is

(A)	$[Co(CN)_6]^{3-} < [Co(NH_3)_6]^{3+} < [Co(NH_3)_5(H_2O)]^{3+} < [Co(NH_3)_5(Cl)]^{2+}$
(B)	$[Co(NH_3)_5(Cl)]^{2+} < [Co(NH_3)_5(H_2O)]^{3+} < [Co(NH_3)_6]^{3+} < [Co(CN)_6]^{3-}$
(C)	$[Co(CN)_6]^{3-} < [Co(NH_3)_5(Cl)]^{2+} < [Co(NH_3)_5(H_2O)]^{3+} < [Co(NH_3)_6]^{3+}$
(D)	$[Co(NH_3)_6]^{3+} < [Co(CN)_6]^{3-} < [Co(NH_3)_5(Cl)]^{2+} < [Co(NH_3)_5(H_2O)]^{3+}$

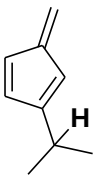
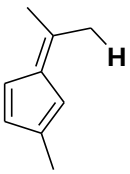
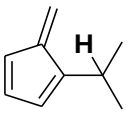
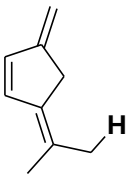
Answer: A

Q.3 One of the products formed from the reaction of permanganate ion with iodide ion in neutral aqueous medium is

(A)	I_2	(B)	IO_3^-	(C)	IO_4^-	(D)	IO_2^-

Answer: B

Q.4 Consider the depicted hydrogen (**H**) in the hydrocarbons given below. The most acidic hydrogen (**H**) is

(A)		(B)	
(C)		(D)	

Answer: B

SECTION 2 (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated **according to the following marking scheme**:

  Full Marks : $+4$ **ONLY** if (all) the correct option(s) is(are) chosen;
  Partial Marks : $+3$ If all the four options are correct but **ONLY** three options are chosen;
  Partial Marks : $+2$ If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
  Partial Marks : $+1$ If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : -2 In all other cases.
- For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then
  choosing ONLY (A), (B) and (D) will get $+4$ marks;
  choosing ONLY (A) and (B) will get $+2$ marks;
  choosing ONLY (A) and (D) will get $+2$ marks;
  choosing ONLY (B) and (D) will get $+2$ marks;
  choosing ONLY (A) will get +1 mark;
  choosing ONLY (B) will get +1 mark;
  choosing ONLY (D) will get +1 mark;
  choosing no option (i.e. the question is unanswered) will get 0 marks; and
  choosing any other combination of options will get -2 marks.

Q.5 Regarding the molecular orbital (MO) energy levels for homonuclear diatomic molecules, the **INCORRECT** statement(s) is(are)

(A)	Bond order of Ne_2 is zero.
(B)	The highest occupied molecular orbital (HOMO) of F_2 is σ-type.
(C)	Bond energy of O_2^+ is smaller than the bond energy of O_2.
(D)	Bond length of Li_2 is larger than the bond length of B_2.

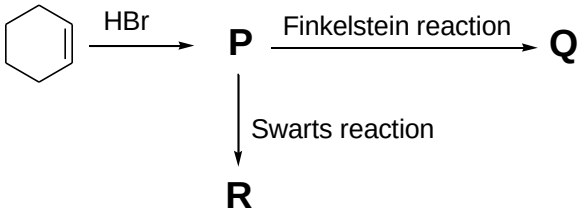
Answer: B, C

Q.6 The pair(s) of diamagnetic ions is(are)

(A)	La^{3+}, Ce^{4+}
(B)	Yb^{2+}, Lu^{3+}
(C)	La^{2+}, Ce^{3+}
(D)	Yb^{3+}, Lu^{2+}

Answer: A, B

Q.7 For the reaction sequence given below, the correct statement(s) is(are)



(In the options, X is any atom other than carbon and hydrogen, and it is different in **P**, **Q** and **R**)

(A)	C–X bond length in **P**, **Q** and **R** follows the order **Q** > **R** > **P**.
(B)	C–X bond enthalpy in **P**, **Q** and **R** follows the order **R** > **P** > **Q**.
(C)	Relative reactivity toward S_N2 reaction in **P**, **Q** and **R** follows the order **P** > **R** > **Q**.
(D)	pK_a value of the conjugate acids of the leaving groups in **P**, **Q** and **R** follows the order **R** > **Q** > **P**.

Answer: B

SECTION 3 (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated **according to the following marking scheme**:

Full Marks : +4 If ONLY the correct numerical value is entered in the designated place;
Zero Marks : 0 In all other cases.

Q.8 In an electrochemical cell, dichromate ions in aqueous acidic medium are reduced to Cr^{3+}. The current (in amperes) that flows through the cell for 48.25 minutes to produce 1 mole of Cr^{3+} is _____.

Use: 1 Faraday = 96500 C mol^{-1}

Answer: 100

Q.9 At 25 °C, the concentration of H^+ ions in 1.00×10^{-3} M aqueous solution of a weak monobasic acid having acid dissociation constant (K_a) of 4.00×10^{-11} is $\boldsymbol{X} \times 10^{-7}$ M. The value of $\boldsymbol{X}$ is _____.

Use: Ionic product of water (K_w) = 1.00×10^{-14} at 25 °C

Answer: [2.2 to 2.3]

Q.10 Molar volume (V_m) of a van der Waals gas can be calculated by expressing the van der Waals equation as a cubic equation with V_m as the variable. The ratio (in mol dm^{-3}) of the coefficient of V_m^2 to the coefficient of V_m for a gas having van der Waals constants a = 6.0 dm^6 atm mol^{-2} and b = 0.060 dm^3 mol^{-1} at 300 K and 300 atm is _____.

Use: Universal gas constant (R) = 0.082 dm^3 atm mol^{-1} K^{-1}

Answer: [-7.2 to -7]

Q.11 Considering ideal gas behavior, the expansion work done (in kJ) when 144 g of water is electrolyzed completely under constant pressure at 300 K is _____.

Use: Universal gas constant (R) = 8.3 J K^{-1} mol^{-1}; Atomic mass (in amu): H = 1, O = 16

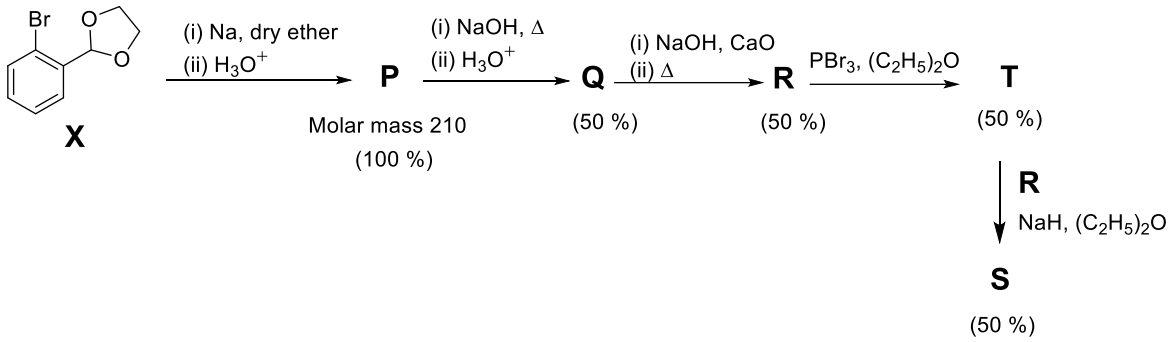
Answer: [-29.95 to -29.8] OR [29.8 to 29.95]

Q.12 The monomer (**X**) involved in the synthesis of Nylon 6,6 gives positive carbylamine test. If 10 moles of **X** are analyzed using Dumas method, the amount (in grams) of nitrogen gas evolved is _____.

Use: Atomic mass of N (in amu) = 14

Answer: 280

Q.13 The reaction sequence given below is carried out with 16 moles of **X**. The yield of the major product in each step is given below the product in parentheses. The amount (in grams) of **S** produced is_____.



Use: Atomic mass (in amu): H = 1, C = 12, O = 16, Br = 80

Answer: 175

SECTION 4 (Maximum Marks: 12)

- This section contains **THREE (03)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists: **List-I** and **List-II**.
- **List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated **according to the following marking scheme**:

Full Marks : +4 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases.

Q.14 The correct match of the group reagents in **List-I** for precipitating the metal ion given in **List-II** from solutions, is

List-I	**List-II**
(P) Passing H_2S in the presence of NH_4OH	(1) Cu^{2+}
(Q) $(NH_4)_2CO_3$ in the presence of NH_4OH	(2) Al^{3+}
(R) NH_4OH in the presence of NH_4Cl	(3) Mn^{2+}
(S) Passing H_2S in the presence of dilute HCl	(4) Ba^{2+}
	(5) Mg^{2+}

(A)	P → 3; Q → 4; R → 2; S → 1
(B)	P → 4; Q → 2; R → 3; S → 1
(C)	P → 3; Q → 4; R → 1; S → 5
(D)	P → 5; Q → 3; R → 2; S → 4

Answer: A

Q.15 The major products obtained from the reactions in **List-II** are the reactants for the named reactions mentioned in **List-I**. Match each entry in **List-I** with the appropriate entry in **List-II** and choose the correct option.

List-I	**List-II**
(P) Stephen reaction	(1) Toluene $\xrightarrow{\text{(i) } CrO_2Cl_2/CS_2 \text{ (ii) } H_3O^+}$
(Q) Sandmeyer reaction	(2) Benzoic acid $\xrightarrow{\text{(i) } PCl_5 \text{ (ii) } NH_3 \text{ (iii) } P_4O_{10}, \Delta}$
(R) Hoffmann bromamide degradation reaction	(3) Nitrobenzene $\xrightarrow{\text{(i) Fe, HCl (ii) HCl, } NaNO_2 \text{ (273-278 K), } H_2O}$
(S) Cannizzaro reaction	(4) Toluene $\xrightarrow{\text{(i) } Cl_2/h\nu, H_2O \text{ (ii) Tollen's reagent (iii) } SO_2Cl_2 \text{ (iv) } NH_3}$
	(5) Aniline $\xrightarrow{\text{(i) } (CH_3CO)_2O, \text{ Pyridine (ii) } HNO_3, H_2SO_4, 288\ K \text{ (iii) aq. NaOH}}$

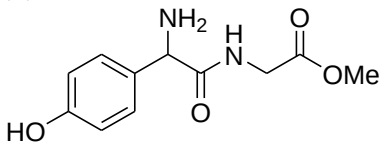
(A)	P $\rightarrow$ 2; Q $\rightarrow$ 4; R $\rightarrow$ 1; S $\rightarrow$ 3
(B)	P $\rightarrow$ 2; Q $\rightarrow$ 3; R $\rightarrow$ 4; S $\rightarrow$ 1
(C)	P $\rightarrow$ 5; Q $\rightarrow$ 3; R $\rightarrow$ 4; S $\rightarrow$ 2
(D)	P $\rightarrow$ 5; Q $\rightarrow$ 4; R $\rightarrow$ 2; S $\rightarrow$ 1

Answer: B

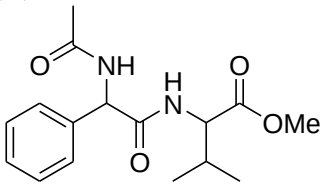
Q.16 Match the compounds in **List-I** with the appropriate observations in **List-II** and choose the correct option.

List-I

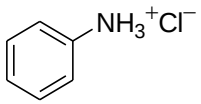
(P)



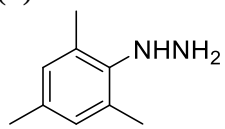
(Q)



(R)



(S)



List-II

(1) Reaction with phenyl diazonium salt gives yellow dye.

(2) Reaction with ninhydrin gives purple color and it also reacts with $FeCl_3$ to give violet color.

(3) Reaction with glucose will give corresponding hydrazone.

(4) Lassiagne extract of the compound treated with dilute HCl followed by addition of aqueous $FeCl_3$ gives blood red color.

(5) After complete hydrolysis, it will give ninhydrin test and it **DOES NOT** give positive phthalein dye test.

(A)	P $\rightarrow$ 1; Q $\rightarrow$ 5; R $\rightarrow$ 4; S $\rightarrow$ 2
(B)	P $\rightarrow$ 2; Q $\rightarrow$ 5; R $\rightarrow$ 1; S $\rightarrow$ 3
(C)	P $\rightarrow$ 5; Q $\rightarrow$ 2; R $\rightarrow$ 1; S $\rightarrow$ 4
(D)	P $\rightarrow$ 2; Q $\rightarrow$ 1; R $\rightarrow$ 5; S $\rightarrow$ 3

Answer: B

END OF THE QUESTION PAPER